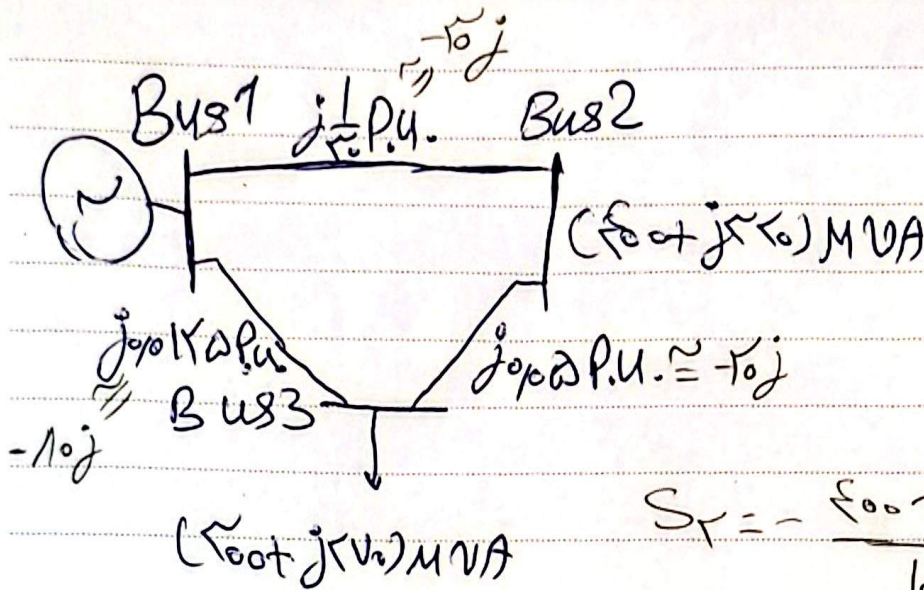


سوال ۵



$$S_r = - \frac{500 + j200}{100} \rightarrow S_r^* = -5 - j2$$

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$$Y = \begin{bmatrix} -1/j & 1/j & 1/j \\ 1/j & -2/j & 1/j \\ 1/j & 1/j & -100/j \end{bmatrix}$$

$$Bus 2 = V_r^{(k+1)} = \frac{1}{Y_{rr}} \left\{ \frac{S_r^*}{(V_r^{(k)})^*} - Y_{r1} V_1 - Y_{r3} V_3^{(k)} \right\} = \frac{1}{-j200} \left\{ \frac{5 + j2}{(V_r^{(k)})^*} - Y_{r1} V_1 - Y_{r3} V_3^{(k)} \right\}$$

$$Bus 3 = V_r^{(k+1)} = \frac{1}{Y_{rr}} \left\{ \frac{S_r^*}{(V_r^{(k)})^*} - Y_{r1} V_1 - Y_{r2} V_2^{(k)} \right\} = \frac{1}{-j100} \left\{ \frac{5 + j2}{(V_r^{(k)})^*} - 1/j - 1/j V_2^{(k)} \right\}$$

مغزی ۱

$$1, \text{ جک } \rightarrow k=0 \rightarrow V_r^0=1, V_r^1=1 \Rightarrow V_r^1 = \frac{-0.045 - j0.1}{1} + 0.4 + 0.9$$

$$= 0.99 - j0.1 \text{ P.u.} \Rightarrow V_r^1 = \frac{-0.025 - j0.05}{1} + 0.1 + 0.1(0.99 - j0.1) = 0.99 - j0.059 \text{ P.u.}$$

$$2, \text{ جک } \rightarrow k=1 \Rightarrow \frac{-0.1045 - j0.1}{0.99 + j0.1} = -0.1045 - j0.109 \rightarrow V_r^2 = 0.99 - j0.109$$

$$\rightarrow V_r^2 = 0.99 - j0.109 \rightarrow \begin{cases} V_r = 0.91 \angle -6.11^\circ \\ V_r = 0.99 \angle -6.11^\circ \end{cases}$$

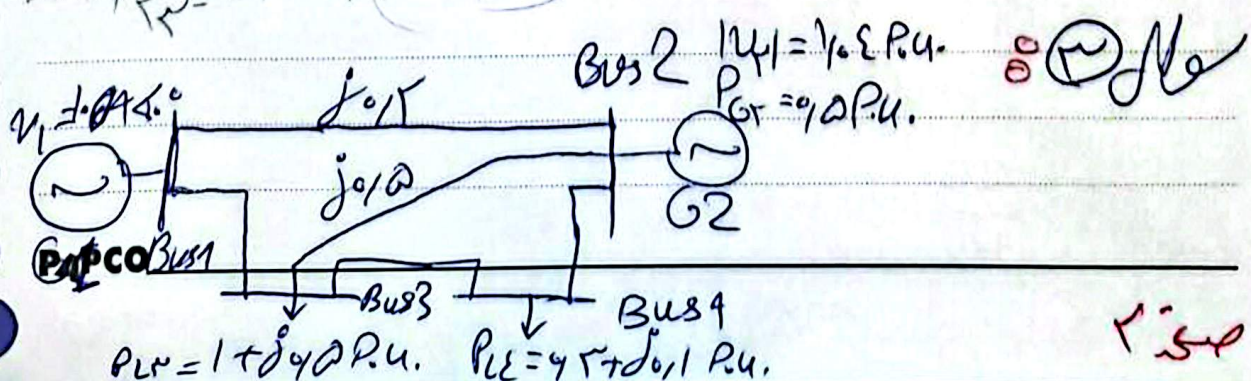
$$S_{ij} = V_i I_{ij}^* = V_i [Y_{ij}(V_i - V_j)]^* \quad \begin{cases} V_r = 0.99 - j0.109 \\ V_r = 0.99 - j0.1 \\ V_1 = 1 \end{cases}$$

$$1, \text{ Bus 1 } \rightarrow P_{1r} = V_1 I_{1r}^* = 1(\epsilon + \epsilon j) \rightarrow P = 2 \text{ P.u.} \rightarrow P = 2 \text{ MW}$$

$$2, \text{ Bus 2 } \rightarrow P_{2r} = V_2 I_{2r}^* = 1(\epsilon + \epsilon j) \rightarrow P = 2 \text{ P.u.} \rightarrow P = 2 \text{ MW}$$

$$3, \text{ Bus 3 } \rightarrow P_{3r} = V_3 I_{3r}^* = (0.99 - j0.1)(-1 - 1j) = -1 - 0.1j \text{ P.u.}$$

$$\rightarrow P_{3r} = -1 \rightarrow P = -1 \text{ MW}$$



جس الف: $Y = \begin{bmatrix} -10j & 0j & 10j & 0 \\ 2j & -9j & 0 & 0 \\ 10j & 0 & -11j & 0 \\ 0 & 0 & 0 & -9j \end{bmatrix}$ $v_1 = 1.6 \angle 0^\circ$
 $P_{v_1} = 1.6, P_{G_1} = \frac{1}{2}, P_{G_2} = 0$
 $P_v = -1, Q_v = -\frac{1}{2}, P_Q = -0.5, Q_Q = -0.1$

جس ب: $v_1^0 = v_2^0 = 1.0 \angle 0^\circ, v_1^0 = 1.0 \angle 0^\circ$

تکرار $\rightarrow v_1^{(k+1)} = \frac{1}{y_{11}} \left\{ \frac{P - jQ}{(v_1^{(k)})^2} - \sum_{2 \neq i} y_{ij} v_i \right\} \Rightarrow v_1^1 = 1.05 \angle 0^\circ$

تکرار $\rightarrow v_1^2 = 1.05 \angle 0^\circ, Q_1 = -0.14$

تکرار $\rightarrow v_1^3 = 1.05 \angle 0^\circ, Q_1 = -0.14, v_2^3 = 1.05 \angle -1.0^\circ$

جس ب: $0.1 \leq Q_{G_1} \leq \frac{1}{2} \rightarrow Q_1 = 0.14$

تکرار $\rightarrow Q_1 = -0.14 \angle -0.1^\circ \rightarrow Q_1 = -0.1, v_1^4 = 0.99 \angle 0.004^\circ$

تکرار $\rightarrow v_1^5 = 0.99 \angle 0.001^\circ = 0.99 \angle 0.001^\circ$

حالا اگر از هر یک در یک مدار $\frac{B}{P}$ و $\frac{B}{P}$ نسبتی پیدا کنیم

$y_{ii} = \sum y_{ij} + \sum \frac{B}{P}$ $N = 2 \rightarrow Y_{2 \times 2}$

$Q_{c,p.u.} = \frac{Q_c}{S_{base}}, y_c = \frac{jQ_c}{120V^2}$

$$I_r = \frac{S_r^*}{V_r^*} = \frac{V_r - V_1}{Z_{1r}}$$

$$V_r^{(k+1)} = V_1 + Z_{1r} \left[\frac{S_r^*}{(V_r^{(k)})^*} \right]$$

$$S_D = 9 \angle 10^\circ \text{ VA}$$

$$\rightarrow S_r = -S_D$$

$$S_r^* = -0.14j \quad C = Z_K S_r^* = (0.02j \times (-0.14j)) = -0.0028$$

$$v_p(k+1) = 1 + \frac{-0.0175 - j0.02}{(v_p(k))^*}$$

$$P_{4PCO} = 1 + \frac{-0.0175 - 0.002}{0.0105} = 0.9145$$

① $V_1 = 1 \angle 0^\circ$ $V_2 = 1 \angle 120^\circ$ $P_2 = 1.2 \text{ p.u.}$ $Q_2 = 0.9 \text{ p.u.}$ $P_1 = 1.2 \text{ p.u.}$ $Q_1 = 0.9 \text{ p.u.}$

$Y = 2 - j$
 $Y = 2 - j$
 $P_2 = -1.2 \text{ p.u.}$
 $Q_2 = +0.9 \text{ p.u.}$
 $S_2 = -1.2 + j0.9 \rightarrow S_2^* = 1.2 - j0.9$
این مقدار را در متنش نداریم

$$Y = \begin{pmatrix} 2 - j & -2 + j & -2 + j \\ -2 + j & 2 - j & 0 \\ -2 + j & 0 & 2 - j \end{pmatrix}$$

د

$$V_2 = V_1 \times (\cos(\delta_2) + j \sin(\delta_2)) + (1.1) \times 2 \rightarrow \cos(\delta_2) + j \sin(\delta_2) = 0.81$$

$$\Rightarrow \delta_2 = 10.1^\circ$$

$S = VI^* \Rightarrow S^* = V^* I \Rightarrow S^* = V_2^* (Y_{21} V_1 + Y_{22} V_2 + Y_{23} V_3)$

$$= V_2^* (-2 + j V_1 + 2 - j V_2) = -2 + j V_1 + (-2 + j) V_2^* V_2 \Rightarrow -1.2 - j0.9$$

$$= 2(1 - j V_1 + (-1 + j) V_2^* V_2) = (2 - j) (1 - V_2^* V_2) \rightarrow \frac{-1.2 - j0.9}{2 - j}$$

$$= |V_2|^2 - V_2^* \Rightarrow V_2 = V_R + j V_I \rightarrow V_2^* = V_R - j V_I \rightarrow |V_2|^2 = V_R^2 + V_I^2$$

P4PCO

$\rightarrow \frac{-1.2 - j0.9}{2 - j} = 0.1044 - j0.1222 \Rightarrow (V_R^2 + V_I^2 - V_R) + j V_I = 0.1044 - j0.1222$

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$$\rightarrow \begin{cases} V_R = 0.95 \text{ V} \\ V_L = 0.125 \text{ V} \end{cases} \rightarrow Z_L = 0.125 \text{ V} \angle 0^\circ \rightarrow |V_L| = 0.125 \text{ V} \text{ p.u.}$$
$$\delta_L = \tan^{-1} \left(\frac{-0.125}{0.95} \right) = -10^\circ$$

$$P_1 = P_{1R} + P_{1L} = V_1 (y_{1R} (V_1 - V_R))^* + V_1 (y_{1L} (V_1 - V_L))^* \quad (d)$$

$$V_L = 1 \angle 10^\circ \text{ V} \Rightarrow P_{1R} = -1/2, P_{1L} = 1/2 \rightarrow P_{\text{total}} = 0.5 \text{ p.u.}$$

$$P_1 + P_2 + P_L = 0.5 + 1/2 - 1/2 = 0.5 \text{ p.u.} \quad (e)$$